Weekly Report – July 27, 2026

Lens Antenna Array

RF & Microwave Lab.

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Sionna-RT 3×3 MIMO Simulation

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Simulation Setup

Sionna-RT Simulation Objective

Objective

Evaluate how the antenna lens changes a 3×3 indoor MIMO channel at 38 GHz using Sionna-RT ray tracing.

- Three TX directions: $+45°$, $0°$, and $-45°$.
- Three RX radiation patterns represent the corresponding lens observation angles.
- With-lens and without-lens channels use identical propagation geometry.

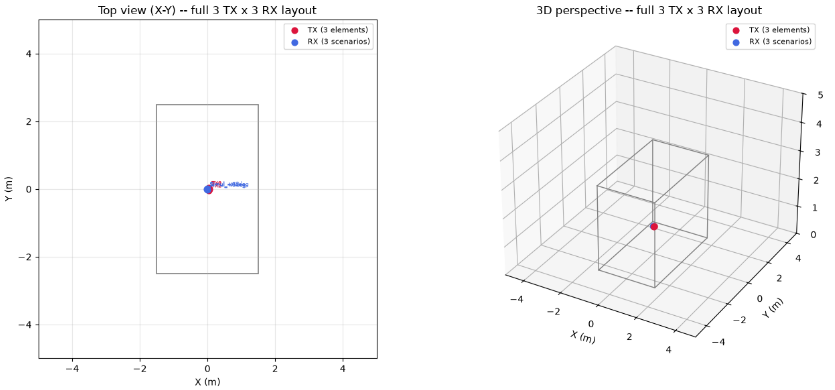
Implementation constraint

Sionna-RT cannot assign a different RX radiation pattern to each RX within the required single run.

Solution

Run three separate 3TX-to-1RX simulations and combine their complex responses into one 3×3 channel matrix.

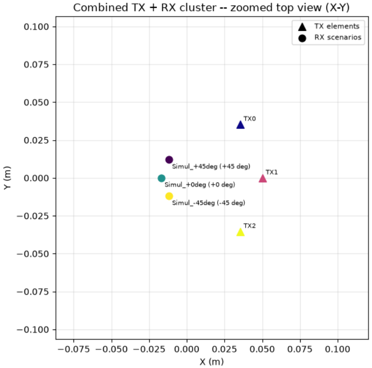
Indoor Simulation Environment: Overall View

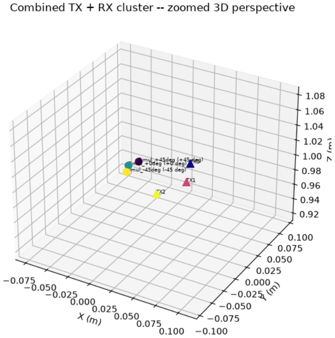


Environment configuration

Room: $3 \times 5 \times 3$ m | Concrete wall: 0.2 m | RX: room center

Indoor Simulation Environment: Detailed View

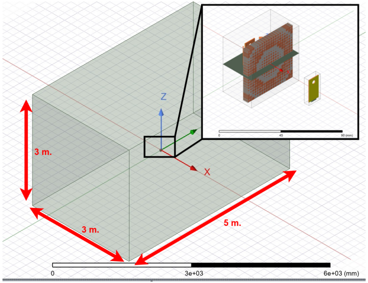




Wideband channel configuration

Center frequency: 38 GHz | Bandwidth: 2 GHz | Range: 37–39 GHz | Samples: 401

Indoor Simulation Environment: HFSS Hybrid Setup



HFSS Hybrid room and close-up of the antenna-lens placement

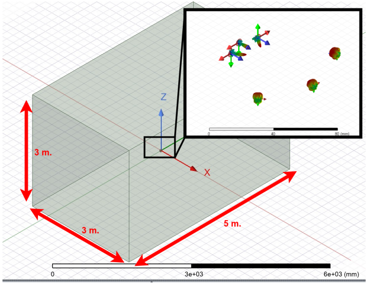
Hybrid representation

- One swept TX source
- Three RX observation points
- Incident angles: $+45°$, $0°$, $-45°$
- With- and without-lens cases

Same physical setup

The room, lens position, focal radius, and RX angular directions follow the same reference geometry used for the Sionna-RT study.

Indoor Simulation Environment: HFSS SBR+ Setup



HFSS SBR+ room and close-up of the physical 3TX-to-3RX arrangement

SBR+ representation

- Three physical TX positions
- Three RX observation points
- Lens and no-lens cases
- Ray-traced response at 37, 38, and 39 GHz

Same physical setup

This is the reference geometry reproduced in Sionna-RT: identical room concept, TX/RX directions, lens placement, and focal

Common Experimental Setup Across the Three Solvers

Clarification

Sionna-RT, HFSS SBR+, and HFSS Hybrid evaluate the same underlying indoor antenna-lens setup.

Common basis	Sionna-RT	HFSS SBR+	HFSS Hybrid
$3 \times 5 \times 3$ m room, 0.2-m concrete walls, centered lens, three angular directions, and with/without-lens comparison	Three 3TX-to-1RX runs are combined into a 3×3 channel	Physical 3TX-to-3RX ray-tracing model	One TX is swept over three incident angles and observed at three RX points

Interpretation

Differences among the figures reflect solver-specific modeling and execution strategies; they do *not* represent different physical room scenarios.

Sionna-RT 3×3 MIMO Simulation

Channel Formulation

Wideband Multipath Channel Model

Frequency response

$$H(f) = \frac{V_{\mathrm{R}}}{V_{\mathrm{T}}} = \sum_{i=1}^{N} \alpha_i e^{-j2\pi f \tau_i}$$

Each path coefficient α_i contains spreading, material interaction, polarization, and antenna-pattern effects; $\tau_i = r_i / c_i$ is its delay.

Complex-baseband impulse response

$$h_{\mathrm{b}}(\tau) = \sum_{i=1}^{N} \alpha_i e^{-j2\pi f_{\mathrm{c}} \tau_i} \delta(\tau - \tau_i)$$

Wideband sampling

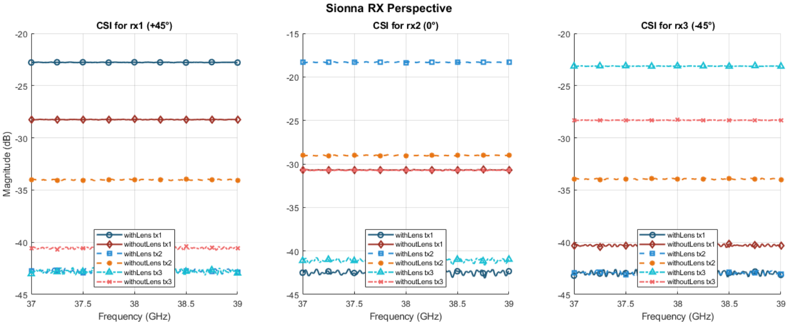
$$\Delta f = \frac{B}{N_f - 1} = \frac{2\ \mathrm{GHz}}{400} = 5\ \mathrm{MHz}$$

- The same 401-point frequency grid is used for all three RX-pattern simulations.
- Complex magnitude and phase are retained when the responses are assembled into $\mathbf{H}(f)$.
- Coherent path summation captures constructive and destructive interference.

Sionna-RT 3×3 MIMO Simulation

Channel-Magnitude Results

Channel Magnitude Across 37–39 GHz



Main observation

The lens strengthens each direction-matched link (TX1–RX1, TX2–RX2, and TX3–RX3), while several non-aligned links are attenuated. The response is nearly flat across the 2-GHz band.

Directional Gain of the Aligned Links

Aligned link	**Lens (dB)**	**No lens (dB)**	**Gain (dB)**
TX1–RX1 $(+45^\circ)$	-22.7	-28.3	$+5.6$
TX2–RX2 (0°)	-18.2	-29.0	$+10.8$
TX3–RX3 (-45°)	-23.0	-28.3	$+5.3$

$$\Delta H_{\text{lens}} = H_{\text{with lens,dB}} - H_{\text{without lens,dB}}$$

Strongest focusing

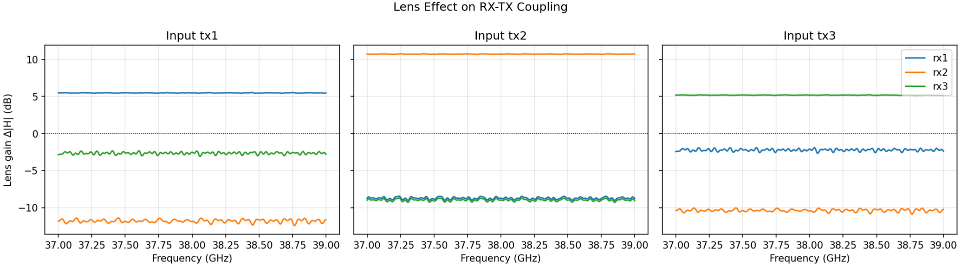
The boresight link TX2–RX2 gains approximately **10.8 dB**.

Angular rejection

For the RX2 perspective, non-aligned TX1 and TX3 are reduced from about -30.5 dB to -42–-41 dB.

The lens reshapes the matrix toward dominant diagonal links instead of providing uniform gain.

Lens-Induced Change in All TX–RX Links

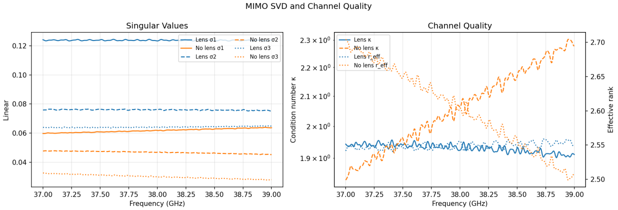


- Aligned links gain approximately $+5.5$, $+10.7$, and $+5.1$ dB for TX1–RX1, TX2–RX2, and TX3–RX3.
- Most cross-directional links are attenuated by approximately 2.5–12 dB.

Sionna-RT 3×3 MIMO Simulation

MIMO Evaluation

Singular Values and Channel Quality



Mean singular values

	Lens	No lens
σ_1	0.1239	0.0618
σ_2	0.0760	0.0467
σ_3	0.0642	0.0300

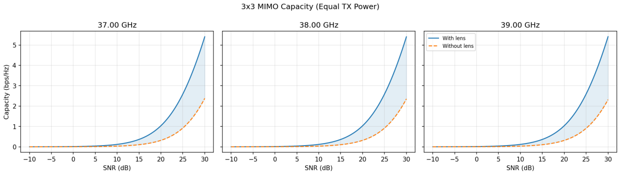
Conditioning

Mean κ: 1.931 with lens versus 2.066 without lens. The lens result is also more stable over frequency.

Qualified rank result

Mean effective rank is 2.549 with lens and 2.603 without lens. The primary benefit is stronger, more stable modes, not a uniformly higher effective rank.

Equal-Power MIMO Capacity



$$C = \log_2 \det\left(\mathbf{I}_3 + \frac{\rho}{3}\mathbf{H}\mathbf{H}^H\right)$$

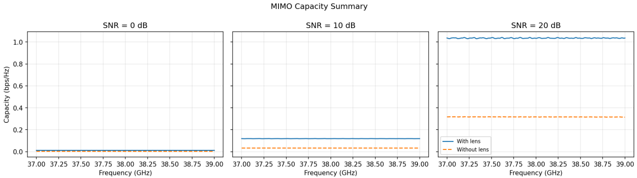
30-dB mean capacity

With lens: 5.406 bps/Hz

Without lens: 2.351 bps/Hz

- Advantage appears at 37, 38, and 39 GHz.
- Small inter-frequency variation confirms a broadband capacity benefit.

Capacity Stability Across Frequency



Full-band mean capacity

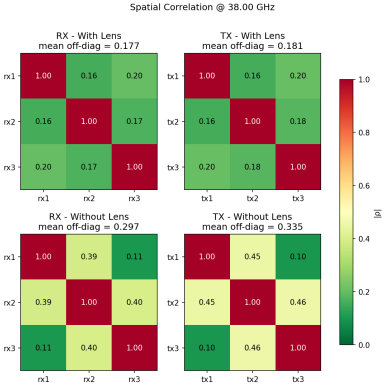
SNR	Lens	No lens
0 dB	0.01211	0.00332
10 dB	0.1191	0.03303
20 dB	1.0353	0.3169

Capacity ratio

Lens/no-lens ratios are approximately **3.65**, **3.61**, and **3.27** at 0, 10, and 20 dB.

Results are nearly frequency-flat over 37–39 GHz.

Spatial Correlation at 38 GHz



Mean off-diagonal correlation

	Lens	No lens
RX	0.177	0.297
TX	0.181	0.335

Reduction

RX correlation decreases by **40.4%**; TX correlation decreases by **46.0%**.

Lower off-diagonal correlation indicates better separation among TX and RX channel vectors.

Sionna-RT 3×3 MIMO Simulation

Conclusions

Sionna-RT Conclusions

What the lens improves

- Strengthens all three direction-matched links.
- Suppresses most cross-directional coupling.
- Increases all three singular-mode strengths.
- Provides stable conditioning and capacity over 37–39 GHz.
- Reduces RX- and TX-side spatial correlation.

Important qualification

The lens does not provide uniform gain and does not increase effective rank at every frequency.

Engineering interpretation

Directional focusing translates into a **stronger and more separable** 3×3 **channel**. The main advantage comes from stronger, more frequency-stable eigenchannels rather than from creating additional spatial modes.

Comparison of Hybrid, Sionna-RT, and SBR+

Comparison of Hybrid, Sionna-RT, and SBR+

Comparison Basis

Three-Solver Comparison Framework

Common comparison basis

- Same underlying antenna-lens geometry
- Canonical $\mathrm{rx1}$–$\mathrm{rx3}$ and $\mathrm{tx1}$–$\mathrm{tx3}$ ordering
- Common frequencies: 37, 38, and 39 GHz
- With-lens and without-lens conditions retained
- Complex magnitude and phase used for MIMO evaluation

Required harmonization

HFSS Hybrid uses swept incident-angle excitations, whereas Sionna-RT and SBR+ use explicit transmitter indices.

Canonical mapping

The native antenna names are remapped into one common 3×3 channel convention before any solver-to-solver comparison is performed.

Raw and Normalized Comparisons Answer Different Questions

Raw complex channel

$$\mathbf{H}(f)$$

Retains absolute propagation gain.

Used for:

- Received channel magnitude
- Absolute lens gain
- Raw MIMO capacity

Frobenius-normalized channel

$$\widetilde{\mathbf{H}} = \frac{\mathbf{H}}{\|\mathbf{H}\|_{\mathrm{F}}}$$

Removes the overall channel-strength difference.

Used for:

- Relative spatial-mode balance
- Normalized singular values
- Structural comparison across solvers

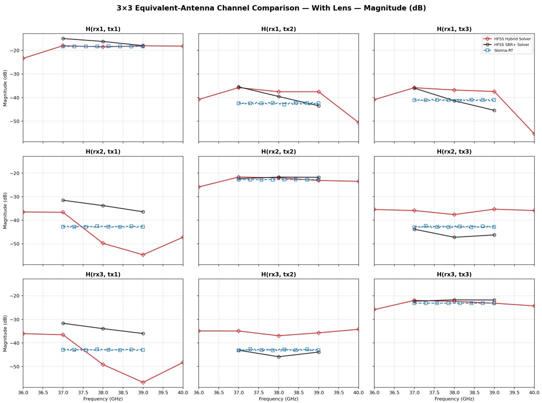
Interpretation rule

Raw and normalized results are complementary and must not be interpreted interchangeably.

Comparison of Hybrid, Sionna-RT, and SBR+

Channel-Magnitude Agreement

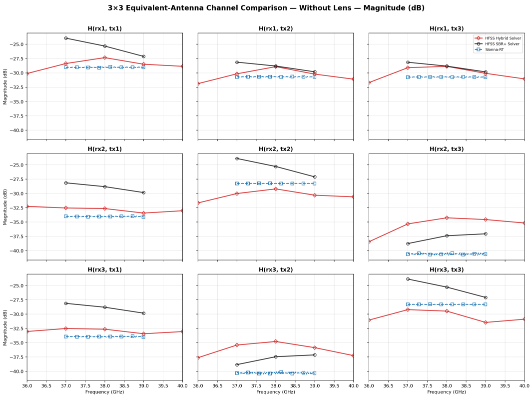
Absolute Channel Magnitude: With Lens



Observation

All solvers reproduce the same broad channel organization, although the predicted absolute magnitudes remain solver-dependent.

Absolute Channel Magnitude: Without Lens



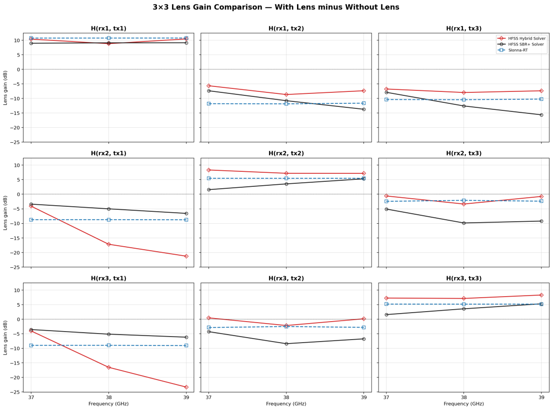
Observation

The common spatial structure remains visible, but differences arise from field formulation, propagation approximation, antenna modeling, and channel extraction.

Comparison of Hybrid, Sionna-RT, and SBR+

Lens-Effect Agreement

Lens-Induced Change Across All Links



$$\Delta H_{\text{lens}} = H_{\text{with,dB}} - H_{\text{without,dB}}$$

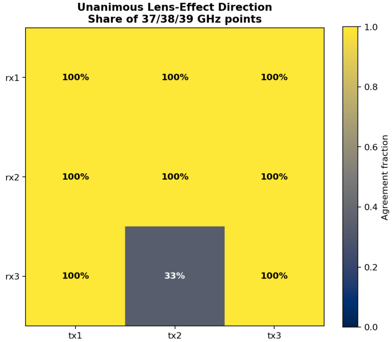
Mean diagonal gain

Hybrid	+8.33 dB
SBR+	+5.33 dB
Sionna-RT	+7.12 dB

Mean off-diagonal change

Hybrid	−7.58 dB
SBR+	−7.86 dB
Sionna-RT	−7.48 dB

Agreement on the Direction of the Lens Effect



Strong qualitative agreement

All three solvers agree on whether the lens increases or decreases magnitude at **25 of 27** link-frequency points:

$$\boxed{92.6\%}$$

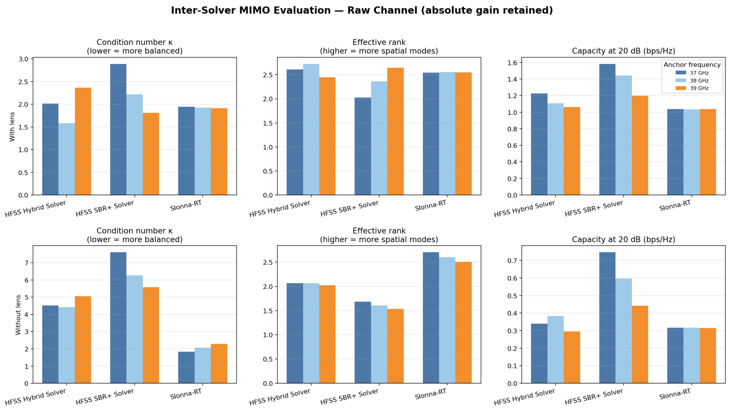
The only non-unanimous cases are $H\,(\mathrm{rx3}, \mathrm{tx2})$ at 37 and 39 GHz.

Exact gain differs, but the direction of the lens effect is highly consistent.

Comparison of Hybrid, Sionna-RT, and SBR+

MIMO Quality and Capacity

Raw-Channel MIMO Evaluation



Common result

At 38 GHz, every solver predicts higher raw 20-dB MIMO capacity after adding the lens.

MIMO Metrics at 38 GHz

Solver	**Lens**	κ	r_{eff}	$C_{20\,\text{dB}}$ **(bps/Hz)**
Hybrid	With	1.579	2.728	1.108
Hybrid	Without	4.424	2.062	0.382
SBR+	With	2.215	2.363	1.443
SBR+	Without	6.255	1.608	0.597
Sionna-RT	With	1.924	2.555	1.033
Sionna-RT	Without	2.064	2.604	0.317

Hybrid

Best with-lens balance: $\kappa = 1.579$ and $r_{\text{eff}} = 2.728$.

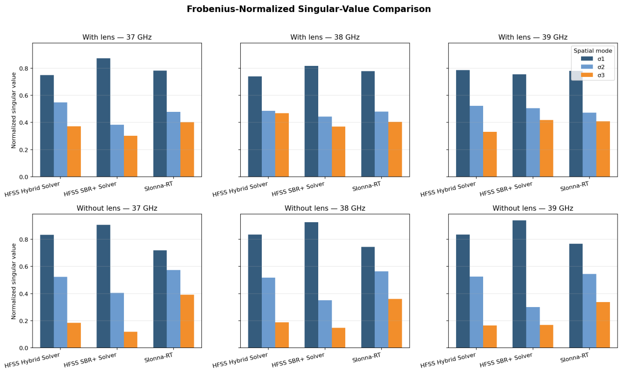
SBR+

Highest raw capacity: 1.443 bps/Hz.

Sionna-RT

Smallest condition-number change, but clear capacity gain from stronger modes.

Normalized Singular-Value Comparison



With lens

All solvers retain three visible spatial modes at the anchor frequencies.

Without lens

Hybrid and especially SBR+ show a relatively weak third mode; Sionna-RT predicts a more balanced normalized distribution.

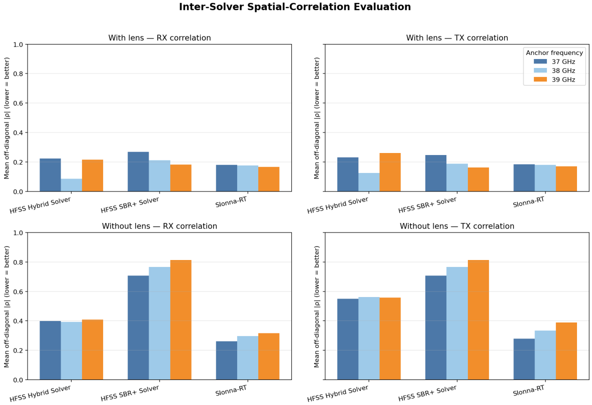
Interpret jointly

Capacity rankings depend on both absolute strength and normalized spatial structure.

Comparison of Hybrid, Sionna-RT, and SBR+

Spatial Correlation

Inter-Solver Spatial-Correlation Comparison



38-GHz mean RX/TX correlation

	Lens	No lens
Hybrid	.085/.126	.393/.561
SBR+	.213/.188	.767/.767
Sionna	.177/.181	.297/.335

Unanimous result

Every solver predicts lower RX- and TX-side spatial correlation after adding the lens.

Comparison of Hybrid, Sionna-RT, and SBR+

Overall Interpretation

What All Three Solvers Agree On

Robust engineering conclusions

The antenna lens:

- Reinforces direction-matched channels
- Suppresses most cross-directional coupling
- Increases raw 20-dB MIMO capacity
- Reduces spatial correlation

Solver-specific strengths

- **Hybrid:** best with-lens spatial balance at 38 GHz
- **SBR+:** highest raw capacity
- **Sionna-RT:** most stable full-band behavior

Boundary of the evidence

Agreement is strongest for qualitative lens trends and relative MIMO improvement. Exact magnitude and phase remain solver-specific until reconciled against measurement.

Monte Carlo 9x9 Position Simulation

Monte Carlo 9x9 Position Simulation

Simulation Design

Monte Carlo Simulation Objective

Question

How does the RX antenna lens affect a 38-GHz indoor channel when nine transmitters are placed at random positions?

- Room size: $20 \times 20 \times 3$ m
- Nine randomly positioned TXs per drop
- Nine RX angular configurations
- With-lens and without-lens cases
- 20 Monte Carlo drops

Paired experiment

Each lens/no-lens comparison uses identical TX positions. This isolates the effect of the RX pattern from random geometry changes.

Evidence status

With only 20 drops and 100,000 path samples per source, the results are preliminary and should be shared with caveats.

Lens and No-Lens Configurations

Without lens

- Same patch RX pattern for every angle label
- RX labels retained only for paired matching
- Expected to provide relatively uniform angular performance

With lens

Different HFSS far-field patterns are used at:

$$+60^\circ, +45^\circ, +30^\circ, +15^\circ, 0^\circ,$$
$$-15^\circ, -30^\circ, -45^\circ, -60^\circ.$$

Performance can therefore depend strongly on angular alignment.

Dataset size

Each notebook contains 20 drops $\times$ 9 RX configurations $=$ 180 drop–RX results; the paired comparison contains 180 pairs.

Simulation Scope and Parameters

Parameter	Value	Parameter	Value
Carrier frequency	38 GHz	Bandwidth	400 MHz
Frequency points	401	Drops	20
TXs per drop	9	RX configurations	9
Random seed	20260718	Path samples/source	100,000
TX height	1.0–2.5 m	Minimum TX–RX range	1.0 m
Wall margin	0.75 m	Power per TX	10 dBm
Noise figure	7 dB	Noise temperature	290 K
Outage threshold	1 bit/s/Hz	Results/notebook	180

TX pattern clarification

Although the notebook filenames contain `Patch`, the Monte Carlo section explicitly forces the TX pattern to isotropic (`iso`).

Random-Position Spatial Coverage

Stored TX-coordinate coverage

Dimension	Range
x	-9.24 to 9.19 m
y	-9.24 to 9.19 m
z	1.01 to 2.50 m

TX–RX distance

Statistic	Distance
Minimum	1.22 m
Mean	10.56 m
Maximum	19.57 m

Coverage check

The 180 stored TX points span a broad portion of the permitted room area and are consistent with the configured sampling constraints.

Monte Carlo 9x9 Position Simulation

Calculation Methodology

Channel and Capacity Calculation

Non-coherent power combination

$$P_{\mathrm{RX}}(f) = P_{\mathrm{TX}} \sum_{i=1}^{9} |H_i(f)|^2$$

Powers from the nine TX channels are added, not their complex voltages.

Equivalent SISO spectral efficiency

$$C = \mathbb{E}_f[\log_2(1 + \mathrm{SNR}(f))]$$

- Noise is calculated from kTB and the configured noise figure.
- Ideal throughput is $B\,C$.
- RMS delay spread is extracted from the IFFT-derived power-delay profile.
- Shannon capacity excludes modulation, coding, overhead, interference, estimation, and RF losses.

Virtual 9x9 MIMO Evaluation

Construction

The nine RX channels associated with different lens angles are combined into a virtual 9×9 matrix.

Reported metrics:

- Condition number
- Effective rank
- Capacity at 10-dB SNR

Interpretation limit

The nine RX states are not measured simultaneously at one physical receiver.

Correct interpretation

These metrics indicate virtual angular-channel diversity and conditioning. They are not directly realizable capacity for a physical single-antenna RX.

Monte Carlo 9x9 Position Simulation

Aggregate Results

Aggregate Lens and No-Lens Results

Metric	No lens	Lens	Change
Median capacity (bit/s/Hz)	7.29	7.67	+0.38 (+5.2%)
5th-percentile capacity	6.48	5.02	−1.46
95th-percentile capacity	9.21	9.68	+0.47
Median throughput (Gbit/s)	2.92	3.07	+0.15 (+5.1%)
Median RMS delay spread (ns)	68.29	65.01	−3.28 (−4.8%)
Outage below 1 bit/s/Hz	0.0%	0.0%	No change

Typical performance

The lens improves median capacity and throughput by approximately **5%**.

Reliability tradeoff

The 5th percentile degrades, indicating greater sensitivity to angle and configuration.

Median Capacity by RX Angle

Angle	No lens	Lens	Difference
$+60°$	7.31	6.70	-0.61
$+45°$	7.31	7.49	$+0.18$
$+30°$	7.31	7.29	-0.01
$+15°$	7.30	8.02	$+0.72$
$0°$	7.30	7.68	$+0.39$
$-15°$	7.29	8.28	$+0.99$
$-30°$	7.29	**8.32**	$+$**1.03**
$-45°$	7.29	8.13	$+0.84$
$-60°$	7.29	7.43	$+0.14$

Best region

The strongest median performance occurs from $-15°$ to $-45°$, led by $-30°$.

Observed asymmetry

$+60°$ performs worse than the no-lens case. Positive and negative angles cannot be assumed symmetric.

Verify RX orientation, HFSS mapping, parameter filtering, and angle signs before treating the asymmetry as physical.

Paired Checks Using Identical TX Positions

Across 180 pairs

Paired statistic	Result
Median capacity difference	$+0.40$ bit/s/Hz
Mean capacity difference	$+0.15$ bit/s/Hz
Lens wins	106/180 (58.9%)
Median SNR difference	$+1.3$ dB
Median delay-spread difference	-4.70 ns

Lens wins by angle

- 15/20 drops at $+15^\circ$ and -15°
- 14/20 drops at 0° and -30°
- 13/20 drops at -45°
- Only 8/20 drops at $+60^\circ$

Conclusion

Lens benefit depends on angular alignment and is not a universal improvement.

Monte Carlo 9x9 Position Simulation

Virtual MIMO Results

Virtual MIMO Channel Conditioning

Metric	**No lens**	**Lens**
Median condition number	49.4 (33.8 dB)	54.0 (34.6 dB)
5th–95th percentile range	27.8–44.4 dB	29.4–40.6 dB
Median effective rank	4.09 of 9	4.07 of 9
Median capacity at 10 dB	21.72 bit/s/Hz	21.11 bit/s/Hz

No virtual-MIMO improvement

The lens slightly increases the condition number, leaves effective rank almost unchanged, and reduces median 10-dB virtual-MIMO capacity by approximately 0.61 bit/s/Hz.

Why this does not contradict the SISO result

A lens can increase received power at selected angles without improving channel orthogonality or the number of independent modes.

Monte Carlo 9x9 Position Simulation

Limitations and Next Steps

Main Limitations and Uncertainty

- Only 20 drops
- 100,000 rather than 1,000,000 path samples
- Isotropic Monte Carlo TX pattern
- 10 dBm applied independently to every TX
- Ideal Shannon-bound capacity
- Lens-angle orientation and mapping require verification
- Virtual, not physical, 9×9 MIMO

Fair-power comparison

For fixed total transmitted power, each TX should be reduced by $10\log_{10}(9) \approx 9.54$ dB.

Monte Carlo 9x9 Position Simulation

Conclusion

Monte Carlo Conclusions

Promising evidence

- Median capacity: $+5.2\%$
- Median throughput: $+5.1\%$
- Median delay spread: -4.8%
- Best angle: -30°

Important qualification

- Lower 5th-percentile capacity
- $+60^\circ$ underperforms no lens
- No virtual-MIMO conditioning improvement
- Strong angle and pattern dependence

Engineering interpretation

The lens behaves as a direction-selective gain mechanism. It can improve typical performance when aligned, but it can be detrimental under angular mismatch. Final design decisions require more drops, fair total power, verified pattern mapping, and physical validation.

Thanks for your attention!